- Chinese remainder theorem (mod 2)

e.g 0011 "3" | "$x+1$"

3×3

```
  0011 x
  0011     (use carries)
  ----
  0011
 0011 +
 -----
 1001
```

$(x+1)(x+1)$ 0011 = 101 = x^2+1

```
   0011 x
   0011      (no carries)
  -----
   0011 +
  0011
  -----
   101
```

can't use polynomial arithmetic

If we must treat the params as integers, then we can sometimes use CRT for efficiency.

Let $M = \prod_{i=1}^{k} m_i$ where the m_i are pairwise relatively prime.

Any integer in $\mathbb{Z}_M$ can be represented by a k-tuple whose elements are in $\mathbb{Z}_{m_i}$.

$A \Leftrightarrow (a_1, a_2, \ldots, a_k)$

where $A \in \mathbb{Z}_M$ and $a_i \in \mathbb{Z}_{m_i}$

1. This mapping is a bijection (a one-to-one correspondance)
   for any A there is a unique k-tuple
   for any k-tuple, there is a unique A
2. Operations performed on elements in $\mathbb{Z}_M$ can be equivalently performed on the corresponding k-tuple elements.

e.g $(A+B) \bmod M \Longleftrightarrow ((a_1+b_1 \bmod m), \ldots, (a_k+b_k \bmod m))$

$A - B$

$A \cdot B$

○ Going back and forth between representations

$A \to (a_1, \cdots, a_k)$: let $a_i = A \bmod m_i \quad 1 \le i \le k$

$(a_1, \cdots, a_k) \to A$

Note that $(a_1, \cdots, a_k) = a_1 \cdot (1, 0, 0, \cdots, 0) + a_2 (0, 1, 0, \cdots, 0) + \cdots + a_k (0, 0, 0, \cdots, 0, 1)$

$\therefore$ we want to find a value $c_1 \in \mathbb{Z}_m$ s.t. $c_1 \Leftrightarrow (1, 0, 0, \cdots, 0)$

i.e. $c_1 \bmod m_1 = 1$, $c_1 \bmod m_j = 0$ for $j \ne 1$

Similarly want to find $c_2 \Leftrightarrow (0, 1, 0, \cdots, 0)$

i.e. $c_2 \bmod m_2 = 1$, $c_2 \bmod m_j = 0$ for $j \ne 2$

$\vdots$

c_k

Recall that $M = m_1 \cdot m_2 \cdots m_k$

Consider the value $M_{\overline{m_1}} = m_2 m_3 \cdots m_k$ (product of all m_s except m_1)

This is almost what we want for c_1 because

$M_{\overline{m_1}} \bmod m_j = 0$ for $j \ne 1$

But $M_{\overline{m_1}} \bmod m_1 = d_1$ (for some non zero value d_1)

$\therefore$ Use extended Euclidean algorithm to find $d_1^{-1} \bmod m_1$

(Because m_1 & $M_{\overline{m_1}}$ are relatively prime)

Now compute $c_1 = M_{\overline{m_1}} \cdot d_1^{-1} = m_2 m_3 \cdots m_k d_1^{-1}$

Clearly $c_1 \bmod m_j = 0$ for $j \ne 1$ and $c_1 \bmod m_1 = d_1 \cdot d_1^{-1} = 1 \bmod m_1$

Do a similar computation for $c_2, \cdots, c_k$

Since $(a_1, \cdots, a_k) = a_1 c_1 + a_2 c_2 + \cdots + a_k c_k$ and all the $c_i \in \mathbb{Z}_m$, then $A = \sum_{i=1}^{k} a_i c_i \bmod M$

Example #1:

$M = m_1 m_2 m_3 = 3 \cdot 5 \cdot 7 = 105$

$M_{\overline{m_1}} = m_2 m_3 = 5 \cdot 7 = 35$, $M_{\overline{m_2}} = m_1 m_3 = 3 \cdot 7 = 21$

$M_{\overline{m_3}} = m_1 m_2 = 3 \cdot 5 = 15$

Now $M_{\overline{m_1}} = 35 \equiv 0 \bmod 5$ and $\equiv 0 \bmod 7$ | $35 \equiv 2 \bmod 3$

$\therefore$ need to find $2^{-1} \bmod 3$ $\therefore$ The value is 2

$\therefore c_1 = M_{\overline{m_1}} \cdot 2^{-1} = 35 \times 2 = 70$

$=$ → Review

$2x \equiv 1 \bmod 3$

$3 = 2 \times 1 + 1$

$1 = \cancel{3} \cdot 1 - 2 \cdot 1$

$2^{-1} \bmod 3 \equiv \cancel{-1} \bmod 3$

$\equiv 2 \bmod 3$

$M_{\overline{m_2}} = 0 \bmod 3$ and $\equiv 0 \bmod 7$ and $\equiv 1 \bmod 5$

$\shortparallel$

21 $\therefore c_2 = M_{\overline{m_2}} = 21$

$M_{\overline{m_3}} = 15 \equiv 0 \bmod 3$ and $\equiv 0 \bmod 5$ $\equiv 1 \bmod 7$

$\therefore c_3 = M_{\overline{m_3}} = 15$

E.g. $98 \rightarrow (98 \bmod 3,\ 98 \bmod 5,\ 98 \bmod 7)$

$= (2, 3, 0)$

$(2,3,0) = [(2 \times 70) + (3 \times 21) + (0 \times 15)] \bmod 105$

$\equiv [140 + 63] \bmod 105$

$\equiv 203 \bmod 105$

$\equiv 98$

$5 = 2\times 2 + 1.$

$1 = 5 - 2\times 2.$

$2^{-1} \bmod 5 \equiv -2 \bmod 5.$

$-2+5=3.$

Example #2:

$M = 5\times 7\times 11 = 385$

$M_{\overline{m_1}} = m_2\cdot m_3 = 7\times 11 = 77$

$M_{\overline{m_2}} = m_1\cdot m_3 = 5\times 11 = 55$

$M_{\overline{m_3}} = m_1\cdot m_2 = 5\times 7 = 35$

$M_{\overline{m_1}} \bmod 5 = 2 \quad \therefore 2^{-1} \bmod 5 = 3$, $\quad M_{\overline{m_1}} \bmod 7 = 0$, $\quad M_{\overline{m_1}} \bmod 11 = 0$

$M_{\overline{m_2}} \bmod 7 = 55 \bmod 7 = 6 \quad \therefore 6^{-1} \bmod 7 = 6$

$6x \equiv 1 \bmod 7$

$x = 6^{-1} \bmod 7$

$7 = 6\times 1 + 1$

$1 = 7 - 6\times 1$

$6^{-1} \bmod 7 \equiv -1 \bmod 7$

$-1+7 = 6.$

$M_{\overline{m_3}} \bmod 11 = 35 \bmod 11 = 2 \quad \therefore 2^{-1} \bmod 11 = 6$

$2x = 1 \bmod 11$

$x = 2^{-1} \bmod 11$

$11 = 2\times 5 + 1$

$1 = 11 - 2\times 5$

$2^{-1} \bmod 11 \equiv -5 \bmod 11$

$-5+11 = 6$

$\therefore c_1 = M_{\overline{m_1}} \cdot 3 = 77\times 3 = 231$

$c_2 = M_{\overline{m_2}} \cdot 6 = 55\times 6 = 330$

$c_3 = M_{\overline{m_3}} \cdot 6 = 35\times 6 = 210$

If our large number is 98:

$98 \to (98 \bmod 5,\ 98 \bmod 7,\ 98 \bmod 11)$

$(3, 0, 10)$

$(3,0,10) \to [(3\times 231) + (0\times 330) + (10\times 210)] \bmod 385$

$= [693 + 2100] \bmod 385$

$= 98$

Do example with 19

Example #2 (cont'd):

If we want $98 \times 19 =$

$(3, 0, 10) \times (4, 5, 8) = [(3\times4) \bmod 5, (0\times5) \bmod 7, (10\times8) \bmod 11]$

$= (2, 0, 3)$

and $(2, 0, 3) \rightarrow [(2\times231) + (0\times310) + (3\times210)] \bmod 385$

$= [462 + 630] \bmod 385$

$= 322$

Note that $98 \times 19 \bmod 385 = 322$

- Powers of an integer, modulo n
    - Primitive root

Recall that for a & n rel. prime, $a^{\phi(n)} \equiv 1 \bmod n$
Euler's theorem where $\phi(n)$ is the Euler's totient function.

More generally, consider $a^m \equiv 1 \bmod n$
- If a & n are relatively prime, then there is at least one such m ($\phi(n)$), but there may be others
- The least positive m for which this holds is called
    - the order of a (mod n)
    - the exponent to which a belongs (mod n)
    - the length of the period generated by a.
- The powers of the base value ("a") form a sequence that is periodic
- The length of this sequence divides $\phi(n)$
- if the length of the sequence is equal to $\phi(n)$, then "a" generates all the non-zero values that are relatively prime to n.
    $\Rightarrow$ "a" is called a primitive root of n

If n is a prime p, then if a is a primitive root of p, then $a, a^2, a^3, \dots, a^{p-1}$ are distinct and form a full set of non-zero residues $\mathbb{Z}_p$
(a is sometimes called a generator of $\mathbb{Z}_p$).
In $\mathbb{Z}_p$, every non-zero element x is a^i for some i

How many exponents i are there? $\phi(p)$ i.e. $p-1$

If we are working mod p, then exponent values are computed mod $\phi(p)$.

Example #3:

We want to compute $98^{19} \bmod 385$.

$$\left[(98 \bmod 5)^{(19 \bmod 4)} \bmod 5,\ (98 \bmod 7)^{(19 \bmod 6)} \bmod 7,\ (98 \bmod 11)^{19 \bmod 10} \bmod 11 \right]$$

$$= \left[3^3 \bmod 5,\ 0' \bmod 7,\ 10^9 \bmod 11 \right]$$

$$= \left[27 \bmod 5,\ 0 \bmod 7,\ (-1)^9 \bmod 11 \right]$$

$$\left(10 \bmod 11 \equiv 10 \equiv -1 \right)$$

$$= [\ 2,\ 0,\ 10\ ]$$

$$(2, 0, 10) \rightarrow [(2 \times 231) + (0 \times 330) + (10 \times 210)] \bmod 385$$

$$= (462 + 2100) \bmod 385$$

$$= 252.$$

Note that $98^{19} \bmod 385 = 252$.

- Discrete logarithms

In ordinary arithmetic, $y = x^i \Rightarrow i = \log_x y$

Now, consider a primitive root "a" for some prime number p..

The powers of a will produce 0 till $p-1$

Any integer b satisfies $b \equiv r \bmod p$ for some $0 \leq r \leq p-1$

$\therefore\ b \equiv a^i \bmod p$ where $0 \leq i < p-1$

"i" is called the discrete logarithm of b to the base a modulo p

$i = d\log_{a,p}(b)$

$d\log_{a,p}(1) = 0$ because $a^0 \bmod p = 1$

$d\log_{a,p}(a) = 1$ because $a^1 \bmod p = a$

$d\log_{a,p}(xy) = [d\log_{a,p}(x) + d\log_{(a,p)}(y)] \bmod \phi(p)$

$$dlog_{a,p}(y^r) = [r \cdot \log_{(a,p)}(y)] \bmod \Phi(p)$$

- Complexity of best-known alg. for finding discrete log

In general, (i.e. mod n where n may not be prime), discrete logs mod n to some base "a" exist only if a is a primitive root of n.

Easy to compute $y = g^x \bmod p$ given g, p & x

At worst perform, $x-1$ repeated multiplications

: $g \times g \times g \times \cdots \times g = g^x$

But computing x given y, g and p seems to be very difficult as the size of p grows

- best known algorithm has complexity $O(e^{(\ln p)^{1/3} (\ln \ln p)^{2/3}})$ which is infeasible for large primes p.